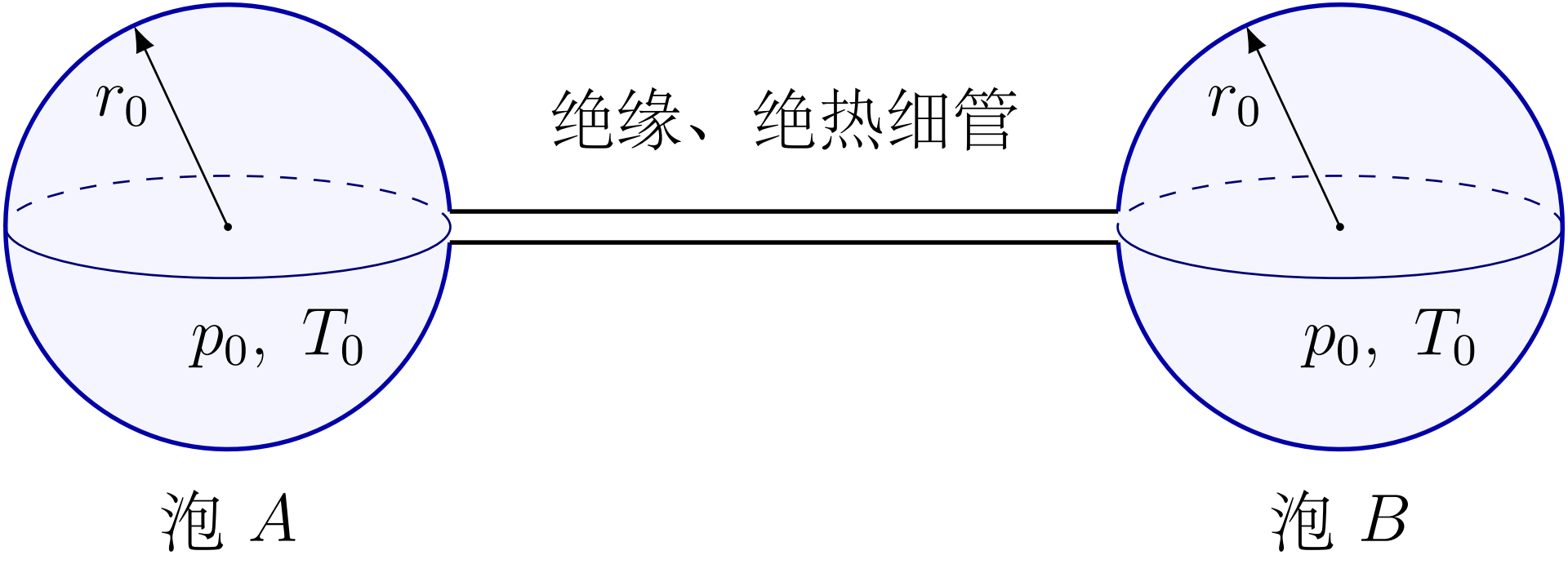
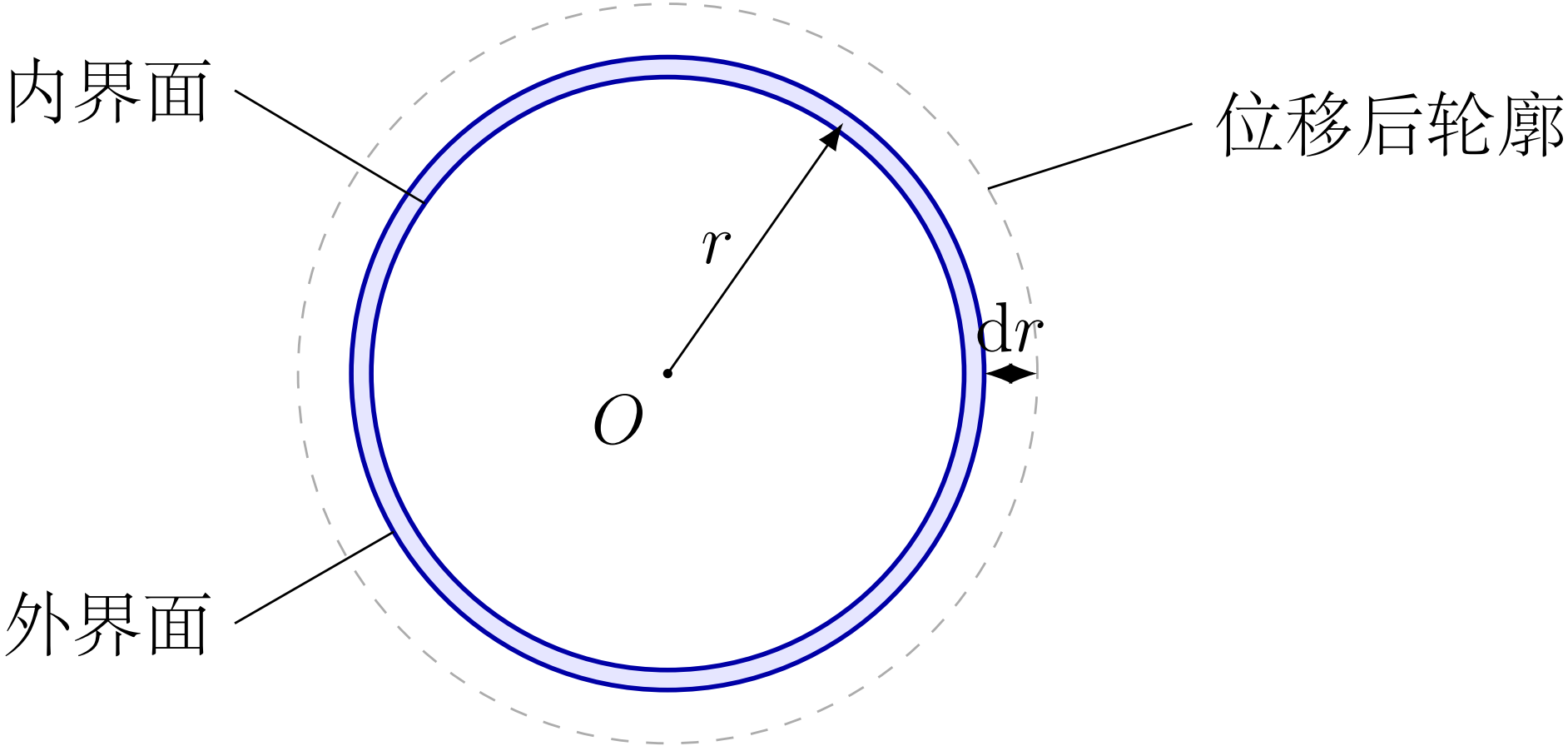
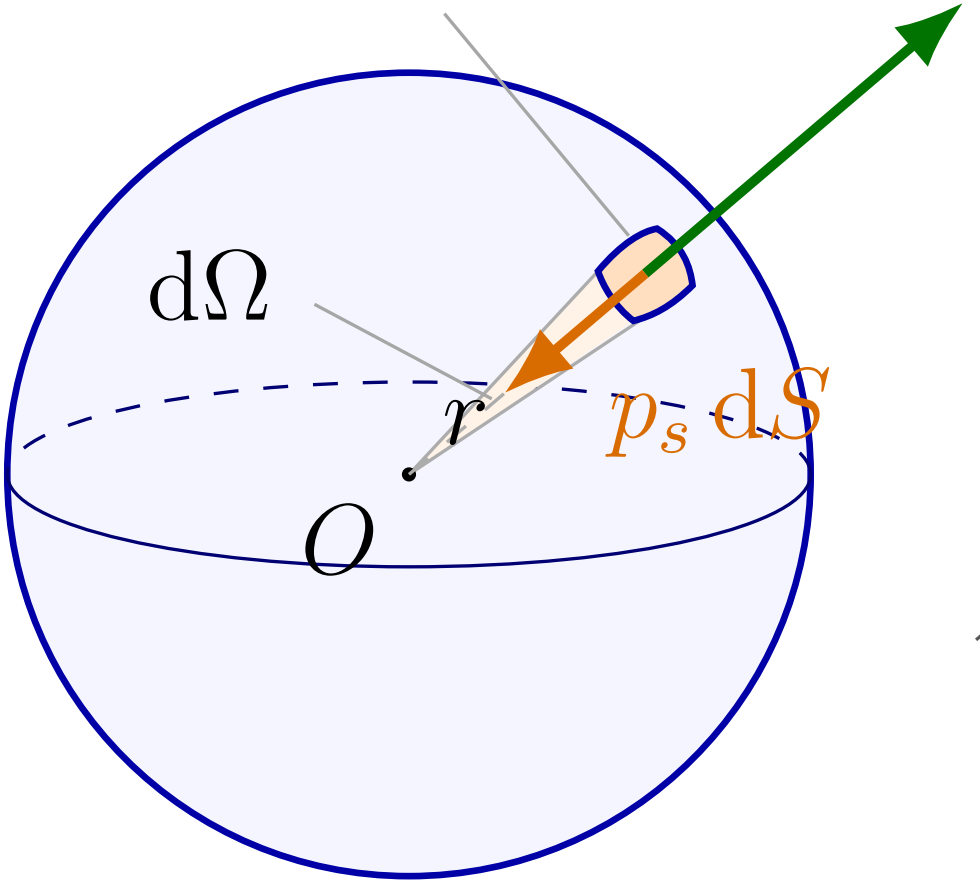

r_0
绝缘、绝热细管
r_0
$p_0,\ T_0$
$p_0,\ T_0$
泡 A
泡 B



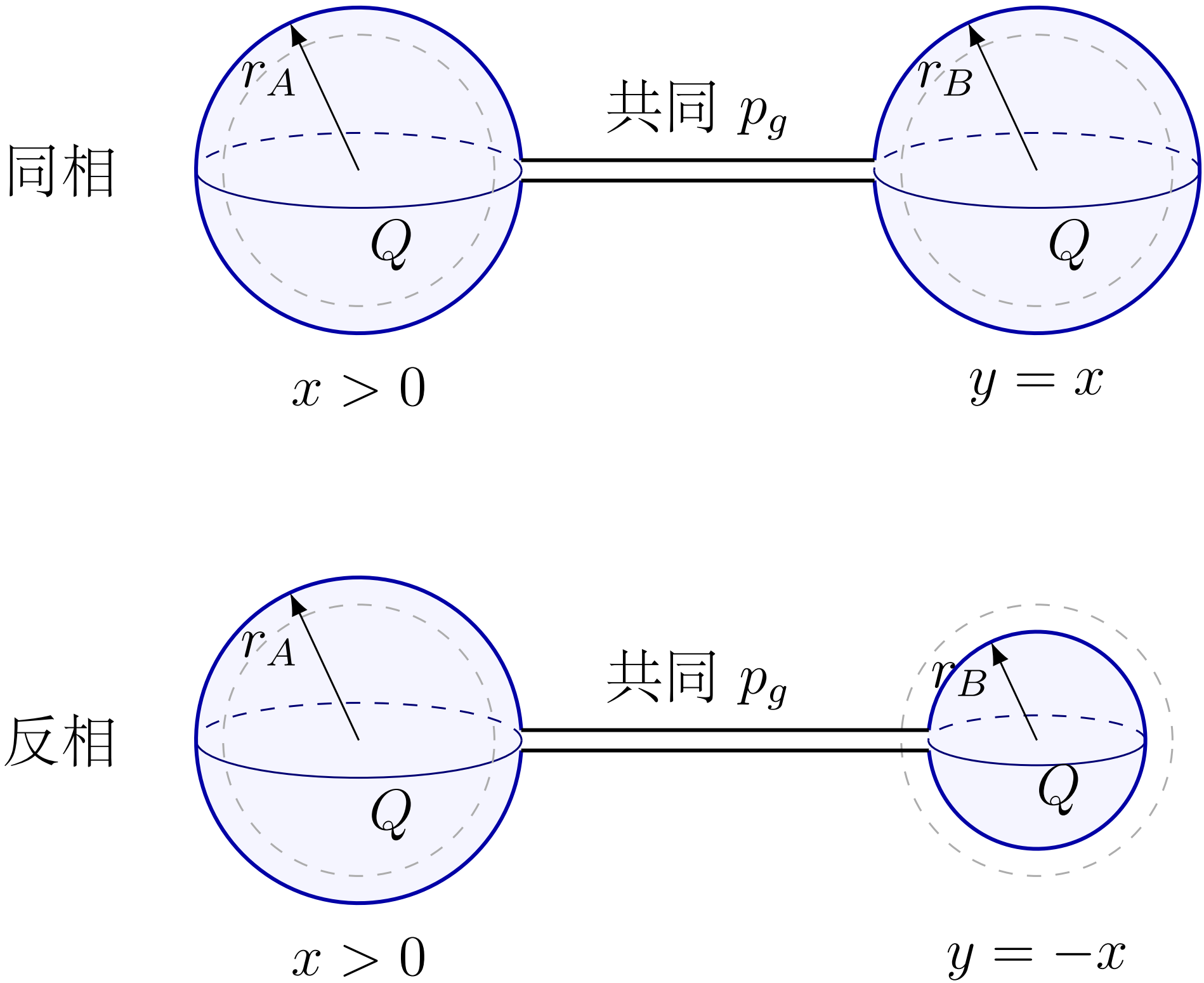
中央截面；膜厚放大显示

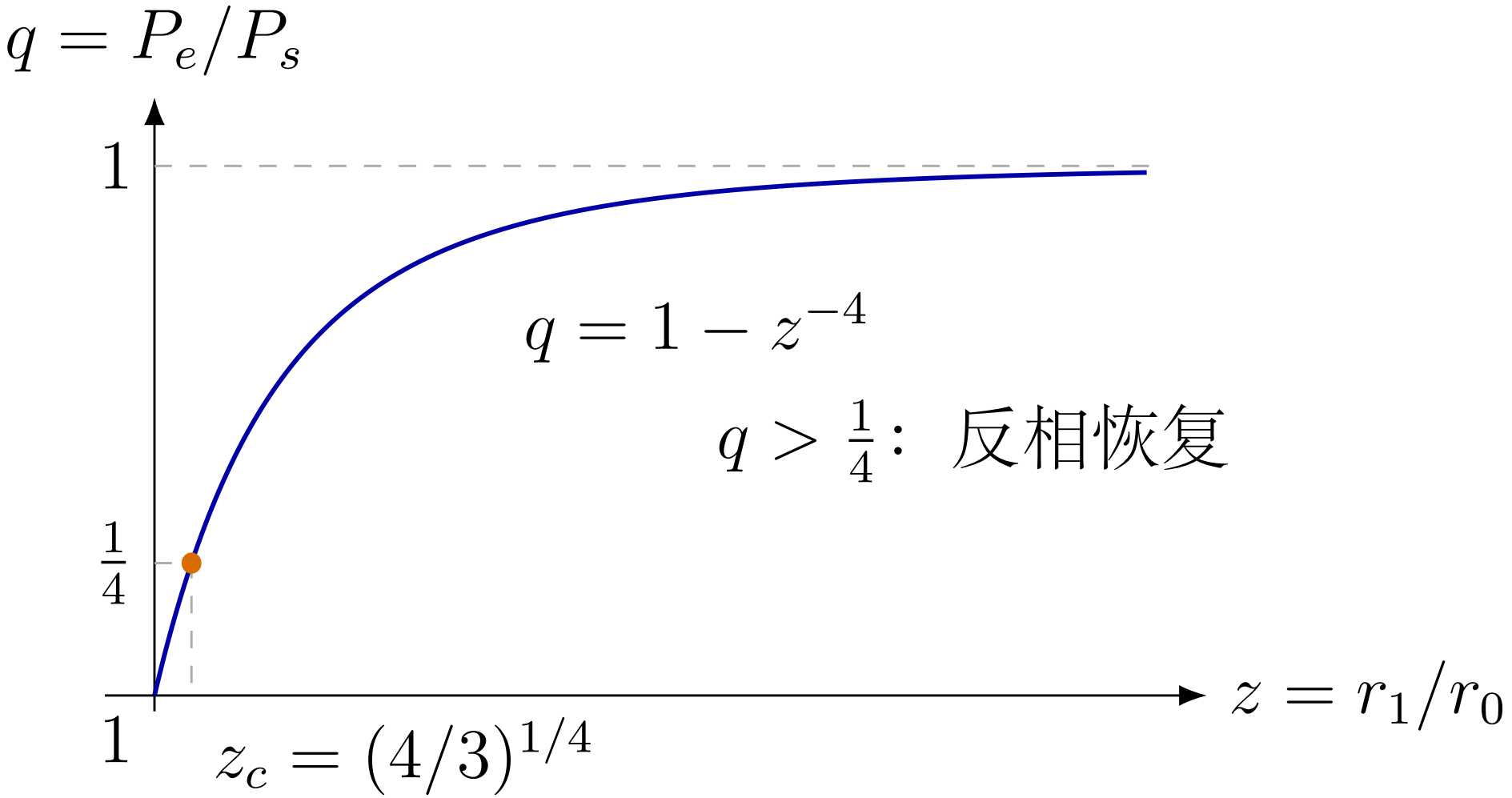
$\mathrm{d}S = r^2 \mathrm{d}\Omega$

$(p_g + p_e)\,\mathrm{d}S$



径向向外


同相
r_A
共同 p_g
r_B
Q
Q
$x > 0$
$y = x$
反相
r_A
共同 p_g
r_B
Q
Q
$x > 0$
$y = -x$
虚线：平衡半径 r_1


$q = P_e/P_s$
1
$q = 1 - z^{-4}$
$q > \frac{1}{4}$: 反相恢复
$\frac{1}{4}$
$z = r_1/r_0$
1
$z_c = (4/3)^{1/4}$